Final ATS Evaluation Report

Day 20 – Final ATS Review & Demo

1. Executive Summary

This report presents the final evaluation of the ATS AI module after completion of development, integration, testing, fairness controls, system evaluation, performance optimization, documentation, and final demonstration activities.

The final review evaluated the major ATS components through regression testing and a controlled live demonstration.

All selected final regression tests completed successfully.

The controlled ATS system evaluation achieved:

- **Accuracy:** 91.67%
- **Precision:** 100.00%
- **Recall:** 87.50%
- **F1 Score:** 93.33%
- **Correct Predictions:** 11 out of 12
- **Mismatch Cases:** 1

A final synthetic live demonstration also successfully showed candidate scoring, ranking, and shortlisting for three candidate profiles.

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2. Evaluation Objectives

The final ATS evaluation was conducted to verify:

1. Semantic candidate-job matching
2. Configurable ATS scoring
3. Candidate ranking
4. Candidate shortlisting
5. Fairness and normalization controls
6. API integration
7. System-level decision consistency
8. Performance and stability
9. End-to-end scoring-to-shortlisting demonstration

The evaluation also aimed to identify known limitations that should remain visible for future improvement.

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3. Final Validation Scope

The following existing test modules were included in the Day 20 regression review:

```
tests.test_semantic_matching_engine
tests.test_ats_scoring_engine
tests.test_candidate_ranking
tests.test_fairness_engine
tests.test_ats_api
tests.test_ats_system_evaluation
tests.test_ats_performance_optimization
```

All seven validation areas completed successfully.

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4. Final Validation Results

Component	Result
Semantic Matching Engine	PASSED
ATS Scoring Engine	PASSED
Candidate Ranking	PASSED
Candidate Shortlisting	PASSED
Fairness & Normalization	PASSED
ATS API Integration	PASSED
ATS System Evaluation	PASSED
Performance & Stability	PASSED

No regression failure was identified during the selected Day 20 validation sequence.

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5. Semantic Matching Evaluation

The Semantic Matching Engine was successfully regression-tested.

The controlled semantic test produced:

Component	Similarity
Skills Similarity	76.63%
Experience Similarity	66.39%
Projects Similarity	84.05%
Overall Similarity	75.69%

Component	Similarity
Match Level	Good Match

The semantic test completed successfully.

The semantic engine also retained the model-reuse and batch-processing optimizations introduced during performance tuning.

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6. ATS Scoring Evaluation

The ATS Scoring Engine was successfully validated using the configured Data Scientist role.

The controlled scoring test used:

Scoring Component	Score
Skill Match	90
Experience Relevance	80
Education Alignment	75
Semantic Similarity	85

The resulting ATS score was:

Overall Score: 85.0

Recommendation: Strong Candidate

This confirmed that the role-based weighted scoring workflow remained operational during final regression testing.

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7. Candidate Ranking and Shortlisting

Candidate ranking and shortlisting were successfully validated using five controlled candidates.

The tested ranking order was:

Rank	Candidate	Score	Decision
1	Candidate A	91	Shortlisted
2	Candidate D	84	Shortlisted
3	Candidate B	78	Review
4	Candidate E	67	Review
5	Candidate C	56	Rejected

The test produced:

- **Shortlisted:** 2
- **Review:** 2
- **Rejected:** 1

The ranking and shortlisting tests completed successfully.

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8. Fairness Evaluation

The fairness test validated controls including:

- Personal attribute masking
- Resume normalization
- Keyword dependency reduction
- Semantic contribution monitoring
- Bias indicator generation

The controlled fairness test reported:

```
Personal Attributes Masked: True
Keyword Dependency Reduced: True
Semantic Contribution Sufficient: True
Bias Flags: None
Requires Review: False
```

The fairness, normalization, and bias-reduction test completed successfully.

These controls support more transparent candidate evaluation but do not replace broader real-world fairness auditing.

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9. ATS API Evaluation

The ATS API regression test successfully validated:

- Health endpoint
- Resume parsing endpoint
- Candidate scoring endpoint
- Candidate shortlisting endpoint
- Request validation
- Job/error handling

The scoring API returned a successful response containing:

- Job role
- Overall ATS score

- Recommendation
- Score breakdown

The API integration test completed successfully.

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10. System Evaluation Dataset

System-level ATS evaluation used **12 controlled candidate cases** covering four categories:

- Tech Role
- Non-Tech Role
- Fresher Resume
- Senior Profile

Automated ATS decisions were compared against predefined manual reference decisions.

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11. Final System Evaluation Metrics

The evaluation produced:

Metric	Result
Total Cases	12
Correct Predictions	11
Incorrect Predictions	1
True Positive	7
True Negative	4
False Positive	0
False Negative	1
Accuracy	91.67%
Precision	100.00%
Recall	87.50%
F1 Score	93.33%
Mismatch Count	1

The results show strong performance on the controlled development dataset while retaining one known false-negative case.

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12. Category-Level Evaluation

Category	Cases	Correct	Accuracy
Tech Role	3	3	100.00%
Non-Tech Role	3	2	66.67%
Fresher Resume	3	3	100.00%
Senior Profile	3	3	100.00%

The strongest results were observed in the Tech Role, Fresher Resume, and Senior Profile categories within this controlled dataset.

The Non-Tech Role category contained the single identified mismatch.

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13. Mismatch Analysis

The mismatch occurred for:

```
Test Case: TC012
Profile: Non-Tech Fresher
Role: Business Analyst
ATS Score: 69.05
ATS Decision: REJECT
Manual Decision: SELECT
```

The candidate score was close to the system-level selection threshold used in the evaluation.

This case demonstrates that borderline candidates can be sensitive to threshold configuration.

Recommended future improvements include:

- Role-specific decision thresholds
- Borderline review zones
- Larger non-tech evaluation datasets
- Fresher-aware calibration
- Human review of near-threshold candidates

The mismatch should remain part of the evaluation record rather than being removed solely to improve the reported metric.

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14. Performance and Stability Evaluation

The final Day 20 performance regression test successfully validated:

- Noisy resume cleaning
- Header/Profile detection
- Resume normalization
- PDF extraction
- Semantic model reuse
- Semantic batch processing
- Memory stability

The final measured run produced:

Performance Measure	Result
Text Cleaning	0.000076 sec
PDF Extraction	0.011058 sec
Semantic Processing	0.066618 sec
Second Model Initialization	0.000855 sec
Peak Tracked Python Allocation	0.0040 MB

These measurements represent the local controlled test environment.

Execution times can vary depending on hardware, model state, caching, document size, runtime environment, and deployment configuration.

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15. Demo Dataset

A synthetic Day 20 demonstration dataset was created for the target role:

Data Scientist

Required skills included:

- Python
- SQL
- Machine Learning
- Pandas
- Scikit-learn

Three synthetic candidates were created to represent:

1. Strong Match
2. Moderate Match
3. Low Match

Synthetic data was used so that the demonstration did not require real candidate personal information.

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16. Live ATS Demonstration

The Day 20 demonstration successfully processed the three synthetic candidate profiles through the existing ATS scoring, ranking, and shortlisting workflow.

The final results were:

Rank	Candidate	Profile	ATS Score	Decision
1	Candidate A	Strong Match	94.60	Shortlisted
2	Candidate B	Moderate Match	68.70	Review
3	Candidate C	Low Match	25.00	Rejected

The demonstration completed with:

```
Total Candidates: 3
Shortlisted: 1
Review: 1
Rejected: 1

ATS live demonstration completed successfully!
```

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17. Demo Methodology Note

The live demo was designed as a controlled integration demonstration.

Skill match was calculated from the overlap between candidate skills and the required Data Scientist skills.

For the demonstration, experience relevance, education alignment, and semantic similarity were supplied as controlled synthetic evaluation inputs representing strong, moderate, and low relevance profiles.

The existing ATS Scoring Engine then calculated the final weighted scores.

The resulting scores were passed to the existing Candidate Ranking Engine and Candidate Shortlisting Engine.

Therefore, the demo validates the integration flow:

```
Demo Candidate Data
        ↓
Evaluation Inputs
        ↓
Existing ATS Scoring Engine
        ↓
Existing Candidate Ranking Engine
        ↓
```

```
Existing Candidate Shortlisting Engine
        ↓
Final Candidate Decision
```

The controlled inputs should not be interpreted as independently predicted values from the full resume-analysis pipeline in this particular demonstration.

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18. Architecture Review

The ATS architecture separates major responsibilities into logical components covering:

```
Resume Processing
        ↓
Cleaning & Normalization
        ↓
Section Classification
        ↓
Candidate Information Processing
        ↓
Skill / Experience / Education Evaluation
        ↓
Semantic Matching
        ↓
ATS Scoring
        ↓
Fairness Controls
        ↓
Ranking
        ↓
Shortlisting
        ↓
API Integration
```

This modular structure supports maintainability, testing, and future extension.

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19. Explainability

The ATS does not return only a final candidate score.

The scoring output includes contributing values such as:

- Skill Match
- Experience Relevance
- Education Alignment
- Semantic Similarity

It also produces a human-readable recommendation.

This makes the candidate evaluation process easier to inspect and troubleshoot.

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20. Final Strengths

The final ATS implementation demonstrates the following strengths:

- Modular architecture
- Resume processing pipeline
- Structured normalization
- Improved PROFILE/header handling
- Semantic candidate-job matching
- Role-configurable scoring
- Explainable score breakdown
- Fairness-related controls
- Candidate ranking
- Automated shortlisting
- Review zone for intermediate candidates
- REST API integration
- Error handling
- Regression testing
- Controlled system evaluation
- Performance optimization
- Developer documentation
- Architecture documentation
- Successful Day 20 live demonstration

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21. Known Limitations

The final review also identified areas that should continue to be improved before large-scale real-world recruitment deployment:

1. The system evaluation dataset is relatively small.
2. Non-tech role validation requires additional cases.
3. One borderline false-negative remains in the controlled evaluation.
4. Role-specific decision thresholds require further calibration.
5. Production-scale load and concurrency testing remain future work.
6. Deployment-level security controls require validation in the target environment.
7. Broader fairness auditing requires more diverse real-world datasets.
8. Human review should remain part of consequential hiring decisions.
9. Performance measurements should be repeated in the actual deployment environment.

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22. Recommended Next Improvements

Future development should prioritize:

- Larger human-reviewed evaluation datasets
- Expanded job-role coverage
- Role-specific threshold calibration
- Borderline candidate review rules
- More extensive non-tech testing
- Production API authentication
- Secure candidate-data handling
- Load and concurrency benchmarking
- Monitoring and observability
- Expanded fairness evaluation
- Automated regression pipelines
- Model and dependency version management

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23. Final Assessment

All selected Day 20 regression tests completed successfully.

The ATS demonstrated successful operation across:

- Semantic matching
- Configurable candidate scoring
- Ranking
- Shortlisting
- Fairness controls
- API integration
- System evaluation
- Performance validation

The controlled system evaluation achieved **91.67% accuracy** and **93.33% F1 score**, with one documented borderline false-negative case.

The Day 20 synthetic live demonstration also completed successfully and produced expected differentiation between strong, moderate, and low candidate profiles.

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24. Final Status

ATS Final Review: COMPLETED

Final Regression Validation: PASSED

Live Demo: PASSED

Demo Dataset: COMPLETED

Technical Documentation: COMPLETED

Known Limitations: DOCUMENTED

The ATS can therefore be presented as the completed AI module required for the internship deliverable, while production-scale deployment should include the additional hardening and validation activities identified in this report.

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25. Conclusion

The final ATS review confirms that the developed system provides a functional and modular candidate-evaluation workflow incorporating resume processing, semantic matching, configurable ATS scoring, fairness-related controls, candidate ranking, shortlisting, and API integration.

The system successfully passed the selected final regression tests and completed the Day 20 controlled live demonstration.

The evaluation also preserves known limitations and the identified borderline mismatch, providing a transparent baseline for future refinement and production hardening.